

Machine Learning in Applications

Detection of Anomalous Behaviour in an Industrial Robot

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CONTENTS

I	Introduction	2
II	Background	2
II-A	Time-series anomaly detection	2
II-B	Autoencoder-based anomaly detection	2
II-C	Adversarial Autoencoder	3
III	Materials and Methods	3
III-A	Dataset	3
III-B	Preprocessing and data split	3
III-C	Windowing	3
III-D	Models	3
III-E	Training and evaluation protocol	4
IV	Results and Discussion	4
IV-A	Main MSE protocol	4
IV-B	Direct AE versus AAE comparison	4
IV-C	Window-length sensitivity	5
IV-D	AAE hyperparameter ablation	5
IV-E	Effect of robust reconstruction loss	5
IV-F	Multi-seed stability	6
IV-G	AAE-specific diagnostics	6
IV-H	Per-feature and per-action analysis	6
IV-I	AAE warm-up with Smooth L1	6
IV-J	Discussion	6
V	Conclusions and Future Works	6
	Appendix A: Additional Figures	8
	References	11

LIST OF FIGURES

1	Example Kuka sensor traces for normal and slow/anomalous behavior. The exact feature names depend on the selected channels, but the figure illustrates the temporal nature of the input data.	4
2	Main precision–recall curves under the official MSE protocol. Isolation Forest achieves the best ranking, while AE and AAE remain substantially below it in PR-AUC.	5
3	AAE ablation over latent dimension and adversarial weight. Hyperparameter tuning did not provide a robust advantage for the AAE.	5
4	Preprocessing and reconstruction-loss ablation. Smooth L1 is the most effective modification for neural reconstruction models.	6
5	Multi-seed stability analysis for the main MSE-protocol models.	7
6	Diagnostic analyses. Feature-level reconstruction errors highlight which channels contribute most to anomaly separation, while per-action differences suggest multimodal robot dynamics.	7

A.1	Test-window count sensitivity for the considered window lengths. This diagnostic supports the decision to exclude overly long windows from the official protocol.	8
A.2	Test anomaly-score distributions for the main MSE-protocol models. The distributions complement the precision–recall curves by showing the overlap between normal and anomalous window scores.	8
A.3	Training dynamics for the best AAE Smooth L1 warm-up configuration. The schedule stabilizes the latent matching process, but this stabilization does not translate into a robust detection gain.	9
A.4	PCA projection of the AAE latent representations by ground-truth label. Anomalous windows are partially shifted with respect to normal windows, but the two classes still overlap considerably. This indicates that the latent space carries some anomaly-related information, but not enough to provide a stronger anomaly score than Smooth L1 reconstruction.	10
A.5	Multi-seed PR-AUC confirmation for the Smooth L1 warm-up experiment. Warm-up does not improve the AAE over the no-warm-up configuration and remains below AE Smooth L1.	10

LIST OF TABLES

I	Dataset composition after loading and action-based segmentation.	3
II	Official split and windowing summary for $w = 64$, stride = 16.	3
III	Main MSE-protocol results on the test set.	4
IV	Direct AAE–AE deltas under the main MSE protocol. Negative values are worse for F1, PR-AUC and ROC-AUC; positive values are worse for false alarms per run.	5
V	Window-length sensitivity. The official protocol uses $w = 64$	5
VI	Effect of Smooth L1 reconstruction loss under $w = 64$, seed 42.	5

Machine Learning in Applications

Detection of Anomalous Behaviour in an Industrial Robot

Abstract—This report investigates anomaly detection in multivariate time-series acquired from an industrial Kuka robot. The goal was to assess whether an Adversarial Autoencoder (AAE), by regularizing the latent space toward a prior distribution, improves over a standard reconstruction-based Autoencoder (AE) for detecting anomalous robot behavior. The raw data were converted into leakage-safe temporal windows by segmenting the original sequences according to robot actions, splitting entire segments into train, validation and test sets, and fitting preprocessing transformations only on normal training data. Several models were compared: PCA reconstruction, Isolation Forest, a multilayer perceptron AE, a convolutional AE, and an AAE. Under the official MSE-based protocol, Isolation Forest achieved the best PR-AUC, while AAE did not improve over AE. Further ablations showed that the main improvement for neural reconstruction models came from replacing MSE with Smooth L1 loss, which made the standard AE competitive with Isolation Forest and substantially reduced false alarms. AAE also benefited from Smooth L1, but remained below the corresponding AE. Additional diagnostics and a warm-up schedule for the adversarial component showed that latent regularization could be stabilized, but did not yield robust anomaly-detection gains. Overall, the results suggest that, for this dataset, robust reconstruction loss is more useful than adversarial latent-space regularization.

I. INTRODUCTION

Industrial robots are increasingly monitored through multivariate sensor streams, including joint positions, velocities, currents and power-related quantities. Detecting deviations from normal behavior is important for maintenance and safety, especially when anomalous behavior appears as a gradual slowdown or loss of precision rather than as a catastrophic failure. This setting is challenging because the signals are temporal, high-dimensional, noisy, and naturally imbalanced: normal behavior is much more frequent than anomalous behavior. The project considered in this report is *Detection of Anomalous Behaviour in Industrial Robot* (2026/AM01). The main research question was not only to obtain a high anomaly-detection score, but to critically evaluate whether adding an adversarial component to an Autoencoder improves anomaly detection on Kuka robot time-series. The central comparison is therefore between a standard Autoencoder (AE), trained to reconstruct normal windows, and an Adversarial Autoencoder (AAE), which adds a discriminator that regularizes the latent code distribution toward a prior. The work makes four main contributions. First, it defines a leakage-safe preprocessing pipeline that segments the raw Kuka recordings into action-based runs, splits complete runs across train, validation and test sets, and creates fixed-length temporal windows. Second, it compares classical and neural anomaly detectors, including PCA, Isolation Forest, AE, convolutional AE and AAE. Third,

it studies the effect of window length, reconstruction loss, preprocessing and AAE-specific hyperparameters. Fourth, it analyzes whether AAE-specific latent and discriminator scores, as well as a warm-up schedule for adversarial training, can recover a useful advantage for the AAE. The final result is critical rather than affirmative: the AAE was successfully implemented and thoroughly evaluated, but it did not provide a robust advantage over a standard AE equipped with an appropriate reconstruction loss. The most effective change was the use of Smooth L1 reconstruction loss, suggesting that robustness to outliers is more important than adversarial latent regularization in this dataset.

II. BACKGROUND

A. Time-series anomaly detection

Time-series anomaly detection aims to identify temporal regions whose behavior is inconsistent with the expected dynamics. In this project, the raw sensor stream is not classified point by point in isolation. Instead, each example is a temporal window containing a short multivariate trajectory. This choice is appropriate because many robot anomalies, such as slowdowns, are not fully identifiable from a single timestamp; they require observing temporal evolution.

Evaluation is particularly delicate in time-series anomaly detection. Metrics based only on point-wise accuracy may be misleading under strong class imbalance, while overly permissive event adjustment can overestimate detection performance [1]. For this reason, the experiments report both threshold-free metrics, such as ROC-AUC and PR-AUC, and threshold-based metrics, such as precision, recall, F1, balanced accuracy, false alarms per run and event recall. PR-AUC is emphasized because the anomaly class is the minority class.

B. Autoencoder-based anomaly detection

An Autoencoder [4] learns an encoder-decoder mapping

$$z = f_{\theta}(x), \quad \hat{x} = g_{\phi}(z), \quad (1)$$

where x is an input window and $\hat{x}$ is its reconstruction. When trained only on normal samples, the model is expected to reconstruct normal windows well, while anomalous windows should yield larger reconstruction error. The anomaly score can therefore be defined as

$$s_{\text{rec}}(x) = \ell(x, \hat{x}), \quad (2)$$

where ℓ is a reconstruction loss, such as MSE or Smooth L1. This approach is simple and effective, but it has two limitations. First, if the Autoencoder has enough capacity, it may

TABLE I
DATASET COMPOSITION AFTER LOADING AND ACTION-BASED
SEGMENTATION.

Source	Rows	Segments
Normal	233792	2340
Slow/anomalous	41538	303

reconstruct anomalous samples too well. Second, the anomaly score is highly dependent on the chosen reconstruction loss. In the Kuka dataset, this second issue proved to be important because some channels exhibit large ranges and outlier-like values.

C. Adversarial Autoencoder

An Adversarial Autoencoder extends the standard Autoencoder by adding a discriminator in the latent space [2]. The discriminator is trained to distinguish samples from a chosen prior distribution, here $p(z) = \mathcal{N}(0, I)$, from latent codes produced by the encoder. The encoder is simultaneously trained to fool the discriminator. The AAE training therefore combines reconstruction and adversarial regularization:

$$\min_{\theta, \phi} \mathcal{L}_{\text{rec}}(x, \hat{x}) + \lambda_{\text{adv}} \mathcal{L}_{\text{adv}}(z). \quad (3)$$

The motivation is that a regularized latent space may improve generalization and provide an additional anomaly signal. However, this assumption is not guaranteed. Robot behavior is likely multimodal, since different actions induce different normal dynamics. Forcing all normal windows toward a single isotropic Gaussian prior may therefore remove useful structure or introduce an overly restrictive constraint.

III. MATERIALS AND METHODS

A. Dataset

The dataset consists of NumPy arrays containing normal and slow/anomalous executions of a Kuka industrial robot. Each row corresponds to one timestep and contains sensor measurements and metadata. After removing non-feature columns such as labels, run identifiers, action and source information, 85 numerical features were used. The original recordings were not treated as two long monolithic sequences. Instead, the pipeline generated artificial run identifiers by segmenting each recording at changes of the robot action. This produced action-homogeneous temporal segments. The resulting dataset composition is shown in Table I.

Figure 1 shows representative normal and slow/anomalous traces. This visualization is useful for interpreting the task as multivariate temporal anomaly detection rather than static tabular classification.

B. Preprocessing and data split

Missing values were handled within each run using interpolation followed by forward and backward filling. The split was performed at run level, not at row or window level. This avoids leakage: two overlapping or temporally adjacent windows from the same action segment cannot appear in both

TABLE II
OFFICIAL SPLIT AND WINDOWING SUMMARY FOR $w = 64$, STRIDE = 16.

Split	Rows	Runs	Windows	Anom. windows	Anom. %
Train	165495	1586	4725	924	19.56
Validation	54623	529	1537	315	20.49
Test	55212	528	1564	320	20.46

training and testing. The scaler was fitted only on normal training rows and then applied to validation and test data. The official protocol used StandardScaler.

After splitting and windowing with the official configuration, the resulting data distribution is reported in Table II. The anomaly proportion is similar across splits, while complete runs remain assigned to a single split.

C. Windowing

Each raw timestep can be represented as $x_t \in \mathbb{R}^{85}$. The models, however, receive temporal windows

$$X_i \in \mathbb{R}^{64 \times 85}, \quad (4)$$

where 64 is the official window length and 85 is the number of features. Consecutive windows are generated with stride 16 and are never allowed to cross run boundaries. A window is labeled anomalous when at least 10% of its internal timesteps are anomalous:

$$y_i = \mathbb{I} \left[\frac{1}{64} \sum_{t \in X_i} y_t \geq 0.10 \right]. \quad (5)$$

Larger windows provide more temporal context but reduce the number of valid windows, especially when runs are short. For this reason, $w = 128$ was excluded from the official protocol and kept only as an exploratory observation. The corresponding test-window count sensitivity is reported as an additional diagnostic in Appendix Fig. A.1.

D. Models

Five model families were evaluated.

PCA. PCA was used as a reconstruction-based baseline. Each window was flattened from 64×85 to 5440 dimensions. PCA was fitted on normal training windows and the anomaly score was the reconstruction error.

Isolation Forest. Isolation Forest [3] was used as a classical non-neural anomaly detector. Instead of using the flattened raw window, each channel was summarized through temporal statistics: mean, standard deviation, minimum, maximum, energy, mean of temporal differences and standard deviation of temporal differences. This produced approximately $85 \times 7 = 595$ statistical features per window.

MLP Autoencoder. The main AE used a flattened input of dimension 5440, hidden layers of size 256 and 128, and a latent dimension of 16. The decoder mirrored the encoder. The anomaly score was the reconstruction error.

Conv1D Autoencoder. The convolutional AE kept the temporal structure by rearranging each input as [channels, time] and applying one-dimensional convolutions along the temporal

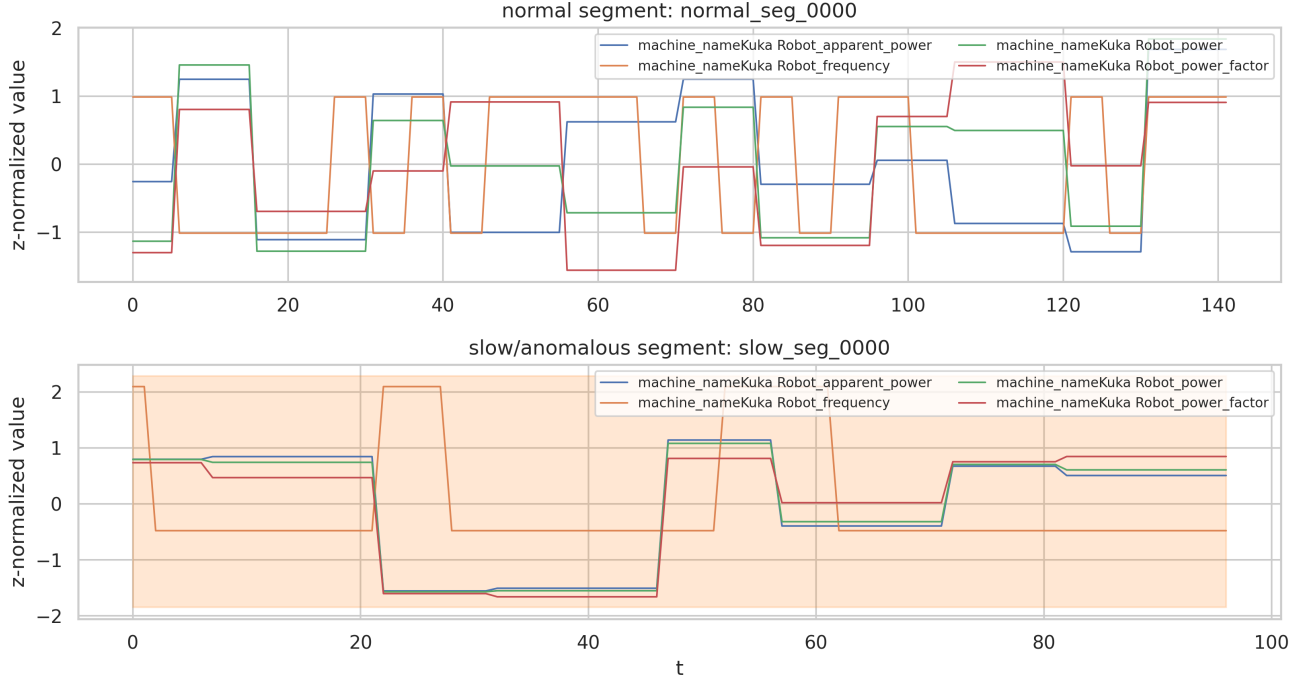


Fig. 1. Example Kuka sensor traces for normal and slow/anomalous behavior. The exact feature names depend on the selected channels, but the figure illustrates the temporal nature of the input data.

axis. This architecture was included to test whether preserving local temporal structure improved over a fully connected AE. **Adversarial Autoencoder.** The AAE used the same MLP encoder–decoder as the AE, but added a latent discriminator trained against samples from $\mathcal{N}(0, I)$. Training alternated reconstruction updates, discriminator updates and adversarial encoder updates.

E. Training and evaluation protocol

Deep models were trained using only normal windows from the training set. Validation data were used for early stopping, threshold selection and model comparison in ablations. The test set was used only for final reporting. Unless otherwise specified, the official protocol used $w = 64$, stride 16, StandardScaler, MSE reconstruction loss and seed 42.

For threshold-free evaluation, ROC-AUC and PR-AUC were computed. For threshold-based evaluation, the threshold was selected on the validation set using best F1 and then applied to the test set. The reported metrics include precision, recall, F1, balanced accuracy, false alarms per run, event recall and detection delay.

IV. RESULTS AND DISCUSSION

A. Main MSE protocol

Table III reports the official MSE-based comparison. Isolation Forest obtained the strongest performance under this protocol, with PR-AUC 0.7196 and F1 0.7156. The two neural reconstruction models based on MSE achieved substantially lower PR-AUC. The AAE was slightly worse than the corresponding AE in all main metrics.

TABLE III
MAIN MSE-PROTOCOL RESULTS ON THE TEST SET.

Model	F1	PR-AUC	ROC-AUC	FA/run
Isolation Forest	0.7156	0.7196	0.9370	0.1499
AE Conv1D	0.6106	0.4572	0.8281	0.1784
AE MLP	0.6109	0.4570	0.8481	0.3017
AAE MLP	0.5797	0.4437	0.8266	0.3397
PCA	0.3934	0.2873	0.6883	0.4459

Figure 2 reports the main precision–recall curves for the official MSE protocol. The curves highlight the advantage of Isolation Forest under MSE, while the score distributions provided in Appendix Fig. A.2 further illustrate the weaker separability of reconstruction-based MSE scores.

B. Direct AE versus AAE comparison

The direct comparison between AE MLP and AAE MLP is the core experiment of the project. Both models use the same windowing, the same MLP encoder–decoder backbone, and the same MSE reconstruction loss. The only conceptual difference is the adversarial latent-space regularization.

The AAE did not improve over the AE. The delta was negative for F1, PR-AUC and ROC-AUC, and positive for false alarms per run, meaning that the AAE produced more false alarms. This is summarized in Table IV.

This result suggests that enforcing a Gaussian latent prior does not automatically improve reconstruction-based anomaly detection. A plausible explanation is that normal robot behavior is multimodal because different actions induce different temporal regimes. A single isotropic Gaussian prior may

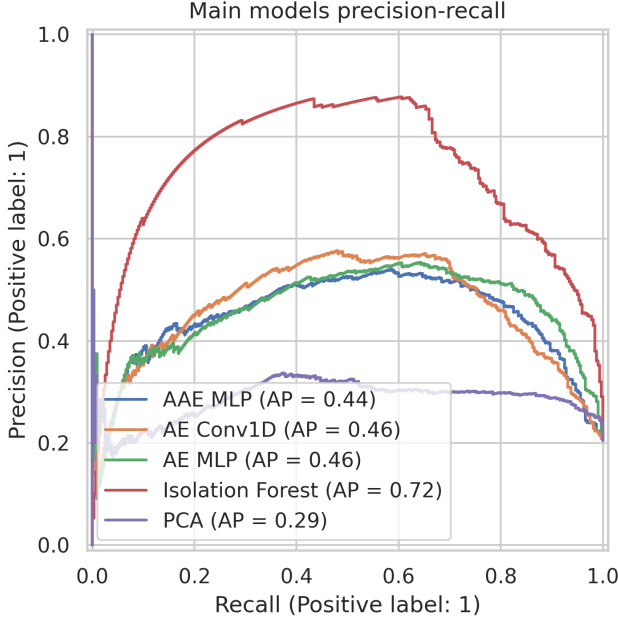


Fig. 2. Main precision–recall curves under the official MSE protocol. Isolation Forest achieves the best ranking, while AE and AAE remain substantially below it in PR-AUC.

TABLE IV

DIRECT AAE–AE DELTAS UNDER THE MAIN MSE PROTOCOL. NEGATIVE VALUES ARE WORSE FOR F1, PR-AUC AND ROC-AUC; POSITIVE VALUES ARE WORSE FOR FALSE ALARMS PER RUN.

Metric	AAE – AE
F1	-0.0312
PR-AUC	-0.0133
ROC-AUC	-0.0216
False alarms/run	+0.0380

TABLE V

WINDOW-LENGTH SENSITIVITY. THE OFFICIAL PROTOCOL USES $w = 64$.

Window	Model	Test windows	PR-AUC	F1
32	AE MLP	2619	0.3787	0.5148
32	AAE MLP	2619	0.3502	0.5058
64	AE MLP	1564	0.4570	0.6109
64	AAE MLP	1564	0.4437	0.5797

regularize the latent space without preserving the structure that is useful for detecting slow anomalies.

C. Window-length sensitivity

Window length materially changes the evaluation problem. Table V compares AE and AAE at $w = 32$ and $w = 64$. Increasing the context from 32 to 64 improves both models, confirming that temporal context is useful. However, AAE remains below AE at both window lengths.

Although exploratory experiments with $w = 128$ produced higher apparent scores, that setting generated too few test windows and a different anomaly prevalence. It was therefore excluded from the official comparison. This prevents the report from over-interpreting a setting that changes the evaluation population rather than only the model behavior.

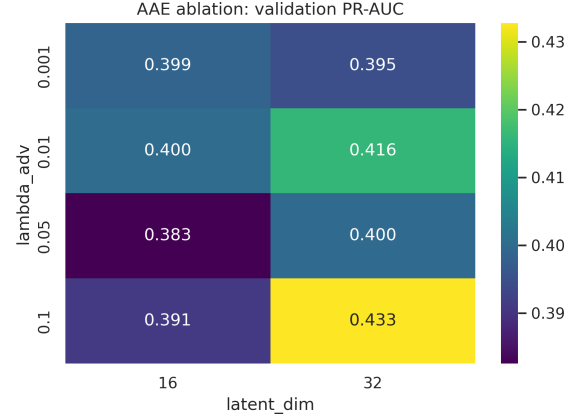


Fig. 3. AAE ablation over latent dimension and adversarial weight. Hyperparameter tuning did not provide a robust advantage for the AAE.

TABLE VI

EFFECT OF SMOOTH L1 RECONSTRUCTION LOSS UNDER $w = 64$, SEED 42.

Model	Loss	F1	PR-AUC	ROC-AUC	FA/run
AE MLP	MSE	0.6109	0.4570	0.8481	0.3017
AAE MLP	MSE	0.5797	0.4437	0.8266	0.3397
AE MLP	Smooth L1	0.7555	0.7634	0.9418	0.1252
AAE MLP	Smooth L1	0.6903	0.7080	0.9035	0.1271

D. AAE hyperparameter ablation

The AAE was further tested by varying latent dimension and adversarial weight. The goal was to verify whether the negative result was caused by an overly strong adversarial loss or by an insufficient latent dimension. The best MSE-based AAE ablation improved over the default AAE only slightly, but did not close the gap to AE with a robust reconstruction loss. Figure 3 summarizes the validation PR-AUC over the tested combinations.

This suggests that the limitation is not only due to a single poor choice of λ_{adv} . The adversarial prior itself appears not to add a sufficiently discriminative signal under the MSE protocol.

E. Effect of robust reconstruction loss

The most relevant improvement emerged from changing the reconstruction loss. MSE penalizes large residuals quadratically and can be dominated by channels with extreme values. Smooth L1 loss behaves quadratically for small errors and linearly for large errors, making it more robust to outliers [5]. In the codebase this option is referred to as `huber`, implemented through PyTorch Smooth L1 loss. Table VI shows that Smooth L1 substantially improved both AE and AAE. AE MLP with StandardScaler and Smooth L1 achieved PR-AUC 0.7634, F1 0.7555 and false alarms per run 0.1252, outperforming the Isolation Forest baseline in the seed-42 official comparison. AAE also improved strongly, but remained below AE. Figure 4 visually confirms that the main gain is associated with the Smooth L1 reconstruction loss, especially when combined with StandardScaler.

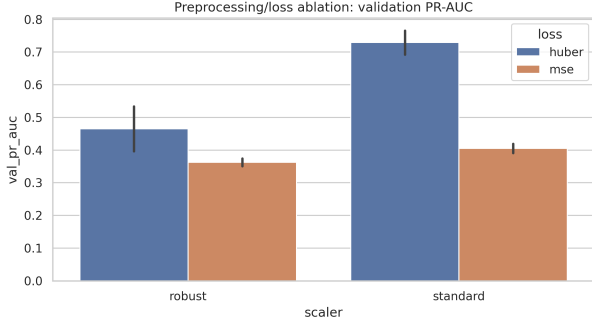


Fig. 4. Preprocessing and reconstruction-loss ablation. Smooth L1 is the most effective modification for neural reconstruction models.

This result changes the interpretation of the project. The initial poor performance of reconstruction models was not an inherent limitation of AE-based anomaly detection; it was largely linked to the choice of reconstruction loss. However, the fact that AE Smooth L1 remains above AAE Smooth L1 indicates that the adversarial component is still not the source of the improvement.

F. Multi-seed stability

The multi-seed experiments evaluated stability across seeds 0, 1 and 2. Under the MSE protocol, Isolation Forest remained the strongest model by mean PR-AUC, while AE stayed above AAE. Figure 5 summarizes this stability analysis for the main MSE-protocol models. In the Smooth L1 warm-up experiment, Isolation Forest kept the highest mean PR-AUC, whereas AE Smooth L1 achieved comparable mean F1 and fewer false alarms. This supports a cautious conclusion: AE Smooth L1 is highly competitive, but the most robust PR-AUC ranking across seeds still favors Isolation Forest.

G. AAE-specific diagnostics

After correcting the diagnostics to use the same reconstruction loss as the selected run, the best AAE-specific score by validation PR-AUC was the Smooth L1 reconstruction score itself. With the corrected diagnostic computation, `score_rec` obtained validation PR-AUC 0.6921 and test PR-AUC 0.7080. Latent-norm, Mahalanobis and discriminator-based scores were therefore useful diagnostic tools, but did not surpass the robust reconstruction score.

This finding is important because it rules out a possible explanation: the AAE is not underperforming simply because it is evaluated with the wrong score. Even when latent and discriminator signals are explicitly inspected, they do not provide a robust advantage over reconstruction.

A two-dimensional PCA visualization of the AAE latent space is reported in Appendix Fig. A.4; it shows partial class separation but also substantial overlap between normal and anomalous windows.

H. Per-feature and per-action analysis

Feature-level diagnostics show that some channels contribute more strongly to the separation between normal and

anomalous windows. This supports the use of robust losses, since a few high-error channels can dominate MSE-based scores. Figure 6 reports feature and action diagnostics derived from the corrected Smooth L1 reconstruction score.

The per-action analysis further suggests that the normal behavior distribution is not unimodal. Different robot actions may form different normal regimes. This observation is consistent with the hypothesis that a single Gaussian prior may be too restrictive for AAE regularization.

I. AAE warm-up with Smooth L1

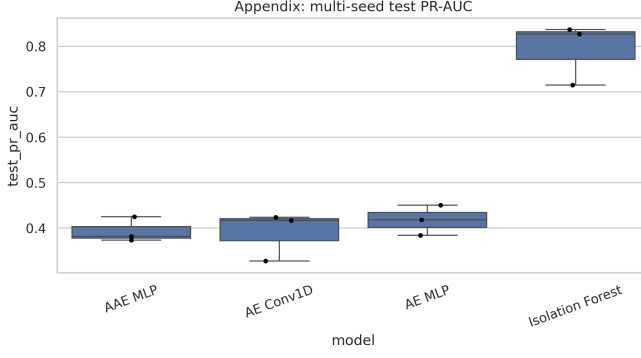
A final experiment tested whether the AAE was failing because the adversarial component was introduced before the encoder-decoder reconstruction had stabilized. The tested strategy used a warm-up phase with reconstruction-only training, followed by a linear ramp of the adversarial weight. The best single-seed configuration used 10 warm-up epochs, 5 ramp epochs and $\lambda_{adv} = 0.01$, obtaining validation PR-AUC 0.7020 and test PR-AUC 0.7028. However, the multi-seed confirmation showed that warm-up did not improve AAE in a robust way. The warm-up AAE remained below AE Smooth L1 by 0.0724 mean PR-AUC and was almost identical to the no-warm-up AAE, with a mean PR-AUC delta of -0.0018. Therefore, warm-up can stabilize latent matching, but does not solve the detection problem. The corresponding training dynamics and multi-seed PR-AUC confirmation are reported in Appendix Fig. A.3 and Appendix Fig. A.5.

J. Discussion

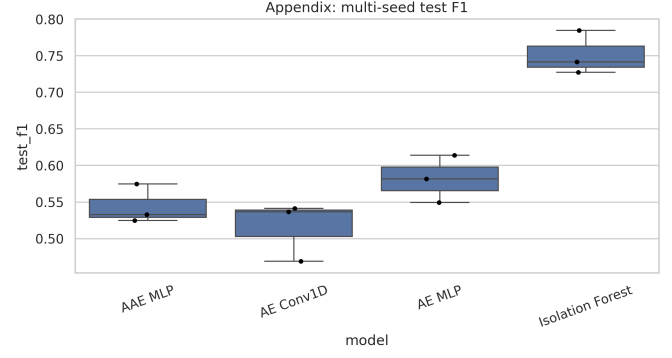
The experiments support three main conclusions. First, the Kuka anomaly-detection task is strongly affected by the representation and scoring choices. The window length changes the evaluation population, and the reconstruction loss changes the ranking quality of neural models. Second, the strongest improvement for deep reconstruction models comes from Smooth L1, not from adversarial regularization. Third, the AAE does not provide a robust advantage over AE, even after hyperparameter tuning, AAE-specific scoring and adversarial warm-up. This does not imply that AAE is generally ineffective. Rather, in this dataset, the normal robot dynamics appear sufficiently complex and multimodal that forcing all latent codes toward a single Gaussian prior does not improve detection. Furthermore, the anomaly type considered here seems to be well captured by robust reconstruction residuals and statistical window descriptors. The extra complexity of the AAE is therefore not justified by the observed performance.

V. CONCLUSIONS AND FUTURE WORKS

This report presented a complete pipeline for anomaly detection in Kuka robot time-series and a critical evaluation of adversarial latent-space regularization. The data were processed as action-homogeneous temporal segments, split at run level to avoid leakage, normalized using only normal training data, and transformed into fixed-length windows. Classical baselines and neural reconstruction models were then evaluated under a consistent protocol.

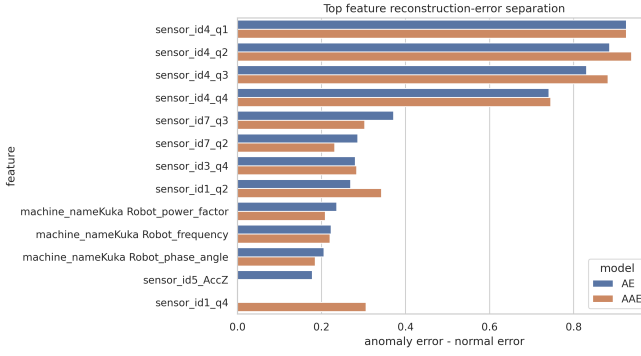


(a) Test PR-AUC stability.

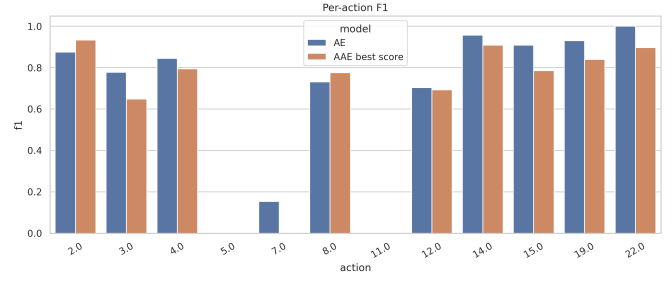


(b) Test F1 stability.

Fig. 5. Multi-seed stability analysis for the main MSE-protocol models.



(a) Top reconstruction-separation features.



(b) Per-action F1.

Fig. 6. Diagnostic analyses. Feature-level reconstruction errors highlight which channels contribute most to anomaly separation, while per-action differences suggest multimodal robot dynamics.

The main findings are as follows. Under the MSE protocol, Isolation Forest was the strongest model, while AAE did not improve AE. Increasing the temporal context from 32 to 64 timesteps improved both AE and AAE, but did not reverse their ranking. Smooth L1 reconstruction loss produced the largest improvement, making AE the best neural model and highly competitive with Isolation Forest. AAE also benefited from Smooth L1 but remained below AE. AAE-specific latent and discriminator scores did not outperform the reconstruction score, and adversarial warm-up did not improve the AAE in multi-seed evaluation. The final answer to the research question is therefore negative but informative: for this Kuka anomaly-detection setting, adversarial regularization is not automatically beneficial and does not justify the added complexity over a standard AE with robust reconstruction loss. Future work should explore conditional or multimodal AAE variants, where the latent prior depends on the robot action or is represented as a mixture of Gaussians. Other promising directions include feature-wise reconstruction weighting, tuning the Smooth L1 transition parameter, evaluating additional anomaly types, and adopting a more detailed event-level protocol aligned with recent recommendations for time-series anomaly detection.

APPENDIX A ADDITIONAL FIGURES

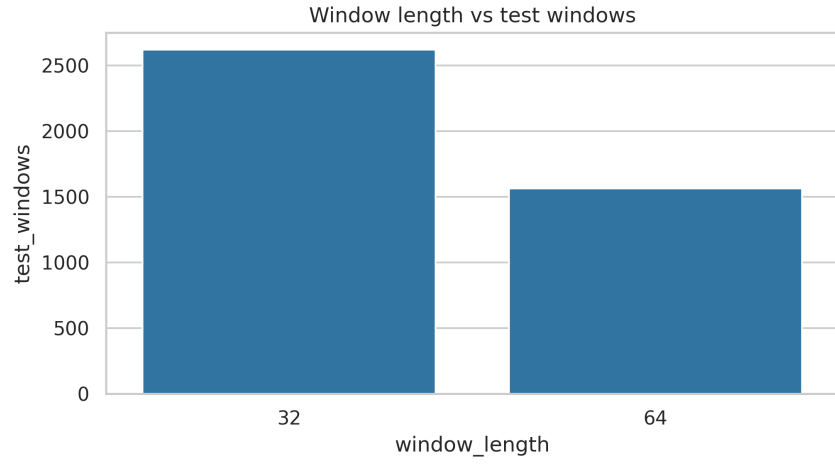


Fig. A.1. Test-window count sensitivity for the considered window lengths. This diagnostic supports the decision to exclude overly long windows from the official protocol.

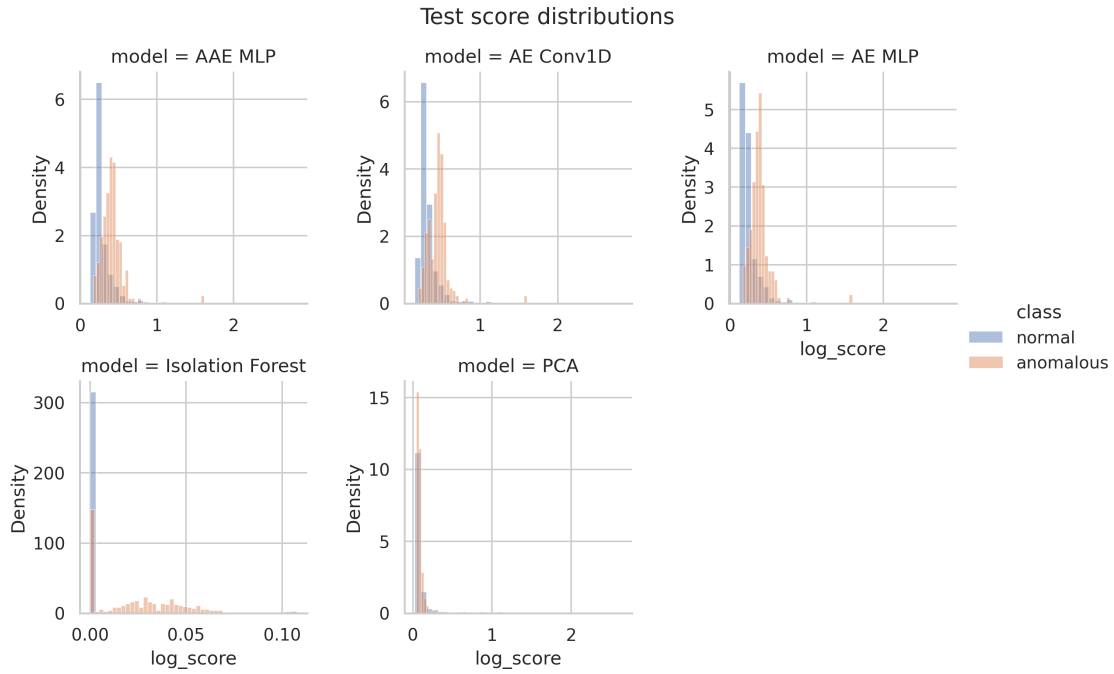


Fig. A.2. Test anomaly-score distributions for the main MSE-protocol models. The distributions complement the precision-recall curves by showing the overlap between normal and anomalous window scores.

Best warm-up AAE training dynamics

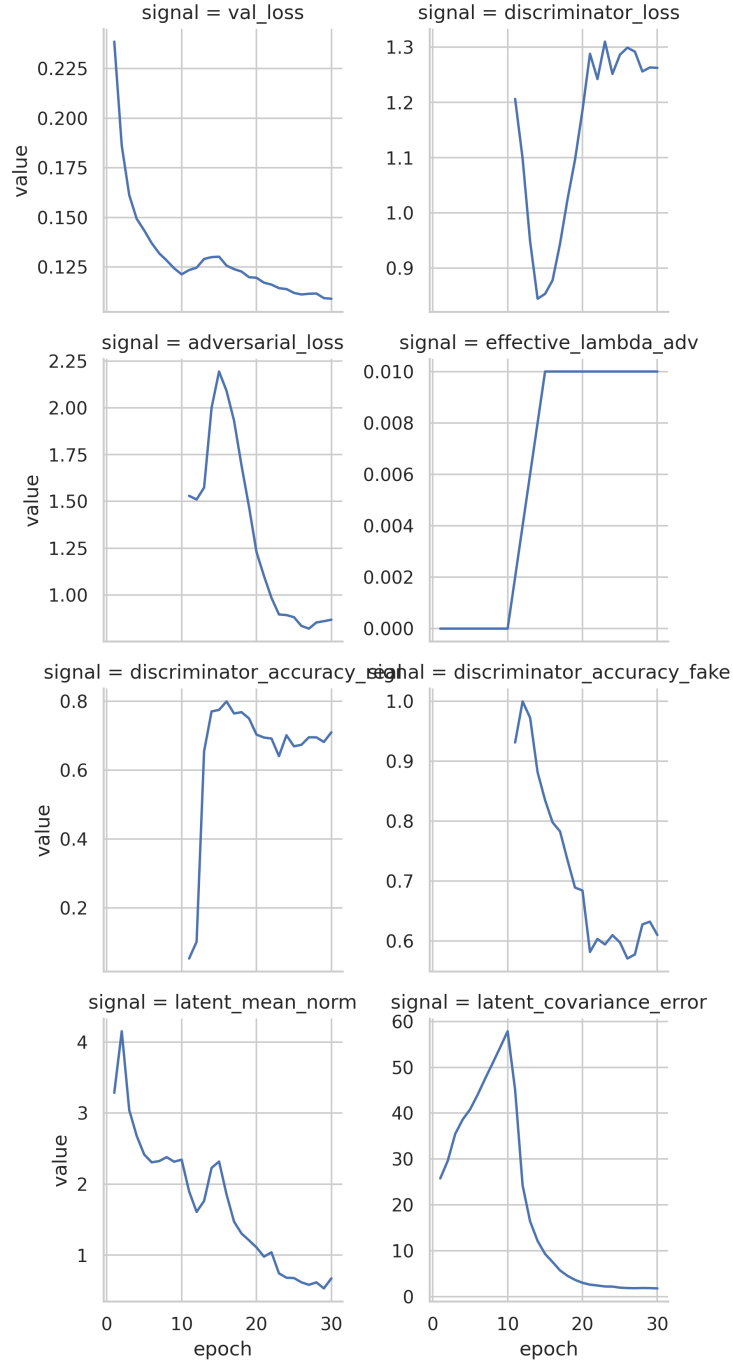


Fig. A.3. Training dynamics for the best AAE Smooth L1 warm-up configuration. The schedule stabilizes the latent matching process, but this stabilization does not translate into a robust detection gain.

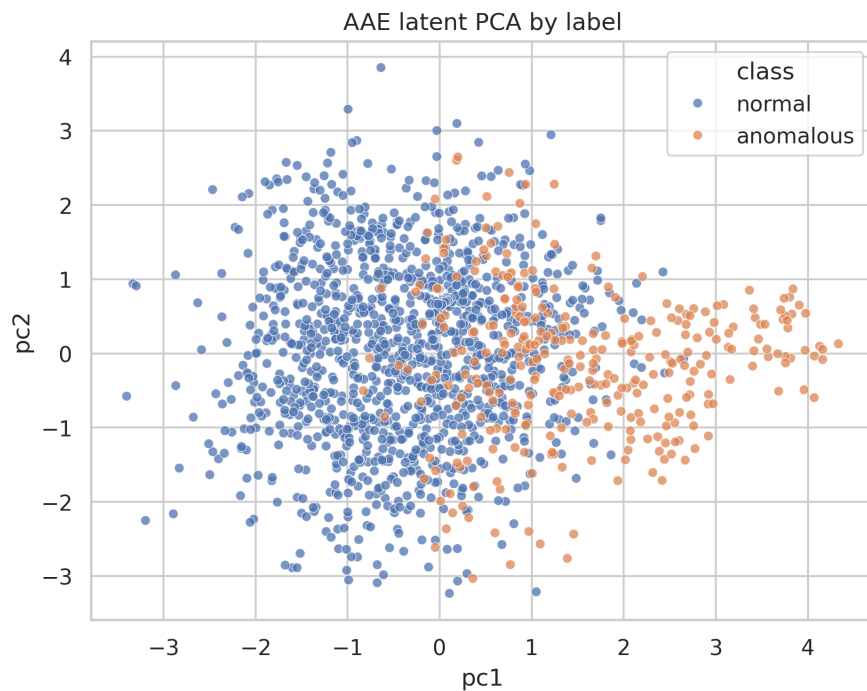


Fig. A.4. PCA projection of the AAE latent representations by ground-truth label. Anomalous windows are partially shifted with respect to normal windows, but the two classes still overlap considerably. This indicates that the latent space carries some anomaly-related information, but not enough to provide a stronger anomaly score than Smooth L1 reconstruction.

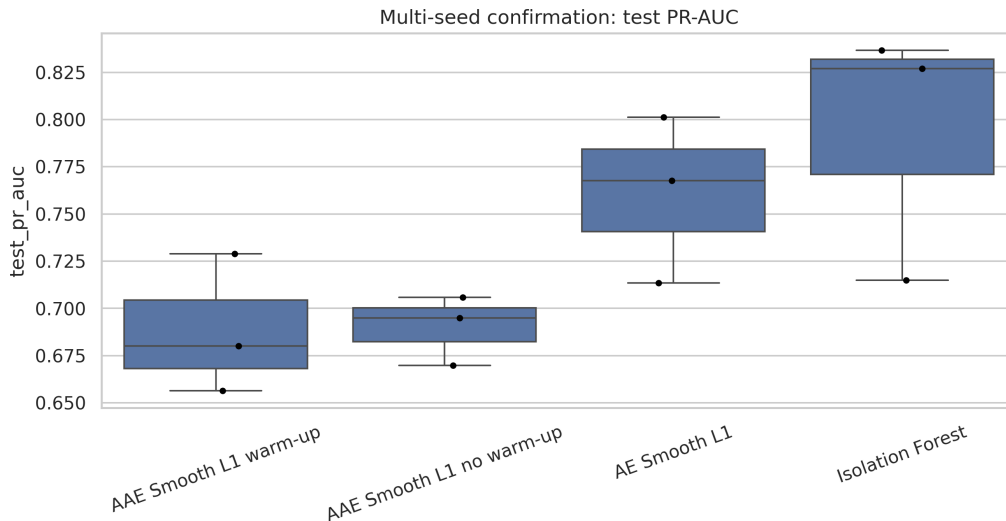


Fig. A.5. Multi-seed PR-AUC confirmation for the Smooth L1 warm-up experiment. Warm-up does not improve the AAE over the no-warm-up configuration and remains below AE Smooth L1.

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